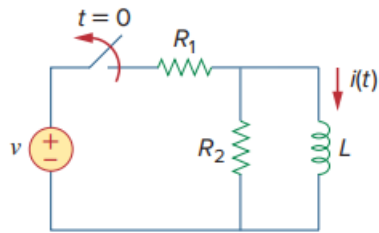


Problem

Using Fig. 7.92, design a problem to help other students better understand source-free RL circuits.

Fig. 7.92:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

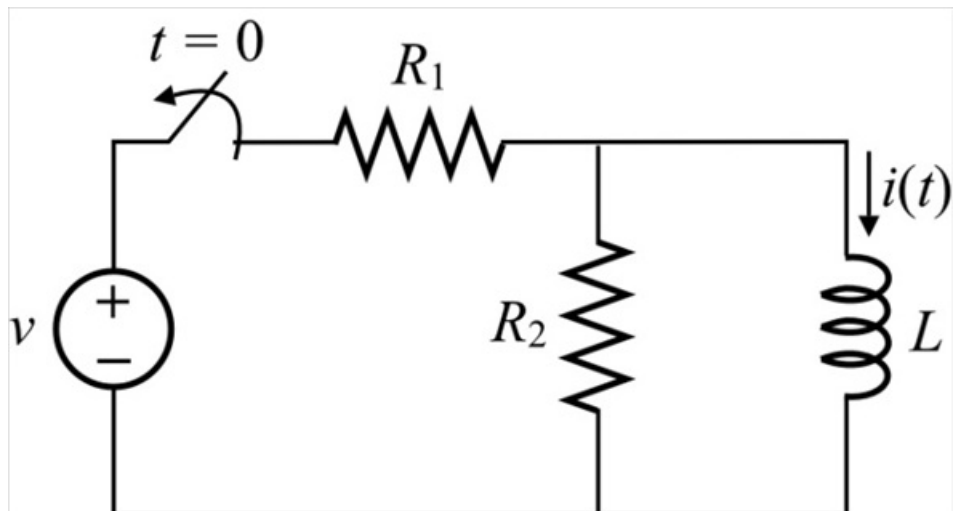


Figure 1

Step 2 of 6

For $t = 0^-$, the switch in the circuit is closed and inductor acts as short circuit. Draw the modified circuit diagram:

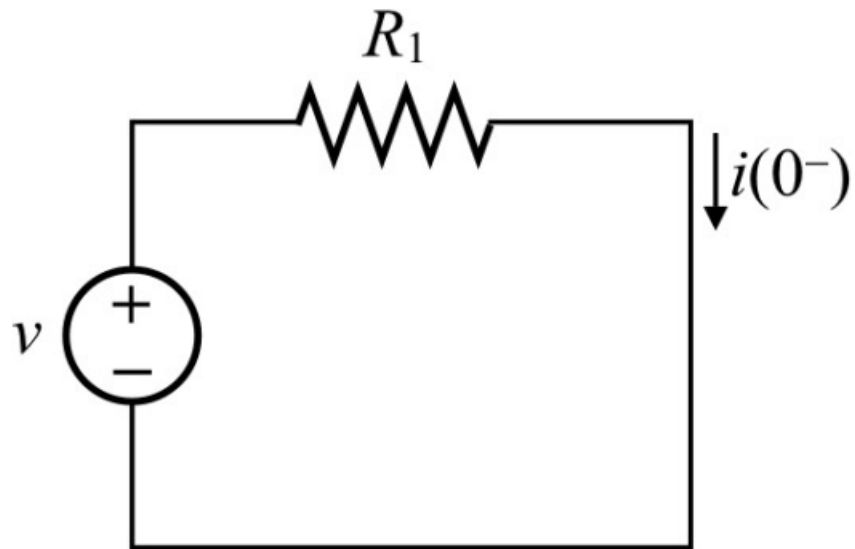


Figure 2

Step 3 of 6

Calculate the initial current through the inductor.

$$i(0^-) = \frac{v}{R_1}$$

Step 4 of 6

For $t > 0$, the switch in the circuit is open. Draw the modified circuit diagram:

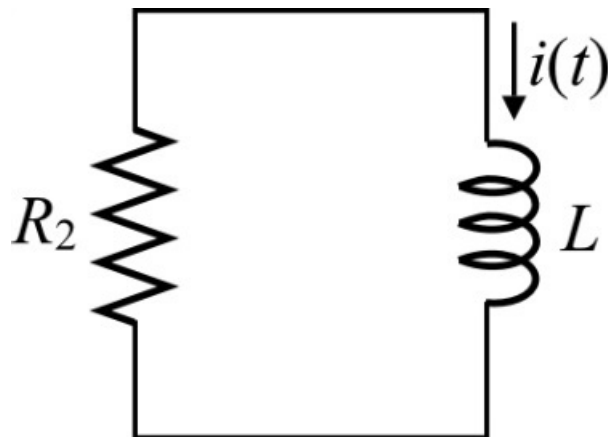


Figure 3

Step 5 of 6

The circuit shown in Figure 2 is a source free circuit. Write an expression for RL source free circuit.

$$i(t) = i(0)e^{-\frac{t}{\tau}} \dots\dots (1)$$

Calculate the time constant of RL circuit at $t > 0$.

$$\begin{aligned}\tau &= \frac{L}{R_{eq}} \\ &= \frac{L}{R_2}\end{aligned}$$

Step 6 of 6

Substitute $\frac{v}{R_1}$ for $i(0)$ and $\frac{L}{R_2}$ for τ in the equation (1).

$$\begin{aligned}i(t) &= \left(\frac{v}{R_1}\right)e^{-\left(\frac{L}{R_2}\right)} \\ &= \frac{v}{R_1}e^{-\frac{R_2 t}{L}}\end{aligned}$$

Hence, the current through inductor $i(t)$ for $t > 0$ is $\boxed{\frac{v}{R_1}e^{-\frac{R_2 t}{L}}}$.